

Problem 5418

Since

$$\frac{k}{k^2 + n^2} = O\left(\frac{1}{n}\right)$$

we can use the expansion

$$e^x = 1 + x + \frac{x^2}{2} + O(x^3).$$

Thus

$$e^{\frac{k}{k^2+n^2}} = 1 + \frac{k}{k^2 + n^2} + \frac{1}{2} \left(\frac{k}{k^2 + n^2} \right)^2 + O\left(\frac{1}{n^3}\right).$$

Insert this into the sum:

$$\begin{aligned} \sum_{k=1}^n e^{\frac{k}{k^2+n^2}} &= \sum_{k=1}^n \left[1 + \frac{k}{k^2 + n^2} + \frac{1}{2} \left(\frac{k}{k^2 + n^2} \right)^2 + O\left(\frac{1}{n^3}\right) \right] \\ &= n + \sum_{k=1}^n \frac{k}{k^2 + n^2} + \frac{1}{2} \sum_{k=1}^n \left(\frac{k}{k^2 + n^2} \right)^2 + O\left(\frac{1}{n}\right). \end{aligned}$$

The quadratic term is $O\left(\frac{1}{n}\right)$:

$$\sum_{k=1}^n \left(\frac{k}{k^2 + n^2} \right)^2 \sim \frac{1}{n} \int_0^1 \frac{x^2}{(1+x^2)^2} dx$$

so it vanishes in the limit. Therefore

$$L = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{k^2 + n^2}.$$

Hence

$$L = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{k/n}{1 + (k/n)^2}.$$

which is the Riemann sum for

$$\int_0^1 \frac{x}{1+x^2} dx.$$

Thus

$$\begin{aligned} L &= \int_0^1 \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2) \Big|_0^1 = \frac{1}{2} \ln 2 \\ \therefore \lim_{n \rightarrow \infty} \left(-n + \sum_{k=1}^n e^{\frac{k}{k^2+n^2}} \right) &= \frac{1}{2} \ln 2 \end{aligned}$$

□